

## Automatic Vehicle Entry Control System Using Image Processing

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### ABSTRACT

The increasing need for efficient and secure access control systems has led to the development of automated vehicle entry systems using image processing. This project focuses on designing and implementing an Automatic Vehicle Entry Control System utilizing image processing techniques to enhance security, reduce manual intervention, and improve operational efficiency. The system captures vehicle images at the entry point, processes them using advanced computer vision algorithms, and performs license plate recognition to grant or deny access based on predefined authorization criteria. This solution is particularly beneficial for gated communities, commercial complexes, and restricted areas, ensuring seamless and automated access control. The implementation of this system significantly reduces human errors, enhances security, and streamlines the vehicle entry process.

**Keywords:-** Vehicle, Image Processing, ANPR

### INTRODUCTION

In modern traffic management, particularly in high-density areas like toll gates, parking facilities, and restricted zones, managing vehicle entry and exit is essential for reducing congestion, enhancing security, and ensuring smooth traffic flow. Traditional manual systems are often slow, labor-intensive, and prone to errors, causing delays and inefficiencies.

Our Image Processing-based Automatic Vehicle Entry Control System addresses these issues by automating vehicle identification and entry control using a Raspberry Pi. Through license plate recognition, the system verifies each vehicle's identity against a database, reducing manual checks and speeding up entry.

This approach supports efficient traffic flow and enhances security with detailed logs for authorized vehicles. At the parking gate, a camera captures each vehicle's image, and Optical Character Recognition (OCR) extracts the license plate number, which is then checked against the database. If there's a match, the Raspberry Pi triggers a DC motor to open the gate for authorized vehicles, while unauthorized vehicles are directed to a separate gate for manual processing.

### LITERATURE SURVEY

1. Pooja Gaikwad et al. (2024) [1] discussed an "Automatic Gate Opening System" aimed at improving security and convenience in controlled-access areas. The study focused on developing a mechanism that detects approaching vehicles and triggers automatic gate

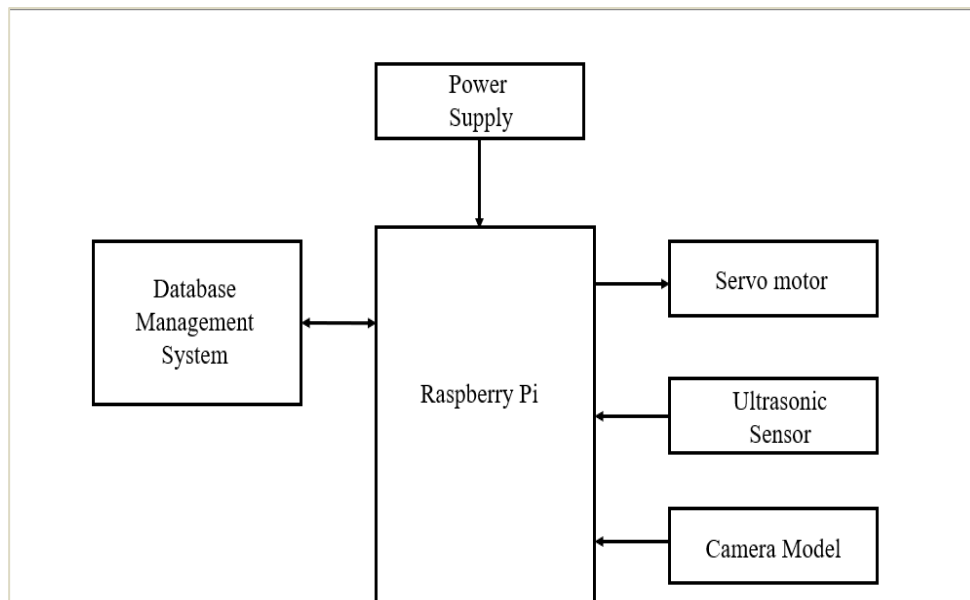
opening, minimizing the need for manual intervention. While the system functioned effectively under test conditions, it faced limitations in accurately identifying vehicles during peak traffic.

2. NI Tushar et al. (2023) [2] proposed a system based on Automatic Number Plate Recognition (ANPR) technology, which successfully detects vehicle number plates and recognizes text through image processing techniques.
3. S. Chandrappa et al. (2022) [3] introduced an IoT-based toll collection system leveraging RFID technology and Raspberry Pi. While the system

streamlined toll collection, it faced challenges related to time complexity. The authors suggest that enhancing automation in toll and vehicle entry-exit control could improve efficiency and reduce traffic congestion, making it a secure and effective method for vehicle management.

4. Saurabh Charde et al. (2020) [4] employed Optical Character Recognition (OCR) with Tesseract and OpenCV libraries to process and recognize vehicle number plates. This integration of machine learning and image processing techniques highlights the potential of OCR in real-time vehicle entry management.

## METHOD



**Fig.1:-Block Diagram**

There is Block-wise description of each component in the Automatic Vehicle Entry Control System, outlining how the system components work together to ensure secure and efficient vehicle access control:

### 1. Raspberry Pi:

The Raspberry Pi serves as the central processing unit of the system, managing data flow between components. It interfaces with the camera module, ultrasonic sensor, and MySQL database

and executes the image processing algorithm for license plate recognition. It processes incoming data and controls access based on verification results.

### 2. Camera Module:

The camera module is connected to the Raspberry Pi and captures real-time images of the approaching vehicle's license plate upon triggering by the ultrasonic sensor. This image is then processed for character extraction, which is essential for identifying and authenticating the vehicle.

### 3. MySQL Database:

The MySQL database stores authorized vehicle details, including license plate numbers and access permissions. After the image processing stage, the extracted license plate number is cross-checked with this database to determine if the vehicle is authorized for entry. If there is a match, the vehicle is granted access.

### 4. Ultrasonic Sensor:

The ultrasonic sensor detects the presence of a vehicle as it approaches the entry gate. It sends a signal to the Raspberry Pi to activate the camera module, ensuring that the system only captures images when a vehicle is present. This sensor reduces unnecessary processing and enhances system efficiency.

### 5. Gate Control Mechanism (Servo Motor):

Upon successful verification of the vehicle's license plate, the Raspberry Pi sends a control signal to the gate's servo motor, allowing the gate to open for authorized vehicles. The servo motor specifications are tailored to handle the gate's weight and range of motion to ensure smooth operation.

### FORMULAE

1.  $\text{Accuracy} = \frac{\text{Correct Predictions}}{(\text{Total Predictions})} \times 100$
2.  $\text{Precision} = \frac{(\text{True Positives})}{(\text{True Positives} + \text{False Positives})} \times 100$

### RESULTS/ DISCUSSION

The Vehicle Entry Control System operates by capturing an image of a vehicle's license plate using the Raspberry Pi Camera. The captured image is processed using an OCR API to extract the license plate number. If the extracted plate number matches a predefined list of authorized vehicles stored in the system, the Raspberry Pi triggers a servo motor to open the gate, granting access.

If the vehicle is unauthorized, access is denied, and the gate remains closed. The system ensures secure and automated vehicle entry using image processing and real-time authentication.

### CONCLUSION

The project has progressed through initial design, component selection, and procurement of essential hardware for the Automatic Vehicle Entry Control System. With all major components, such as the Raspberry Pi, camera module, ultrasonic sensor, and servo motor carefully chosen to meet system requirements, the project is ready for assembly and integration. This preparation ensures that the chosen hardware aligns with the design specifications, allowing for efficient setup in the next phase. Finalizing hardware selection and completing the shopping stage marks a key milestone, laying a strong foundation for testing and implementation.

### FUTURE SCOPE

1. Smart Parking & Traffic Management  
Automated parking lot management with real-time occupancy detection, Integration with smart city infrastructure for dynamic traffic flow control.
2. Integration with Smart Payment Systems - Automated toll and parking fee deduction using digital wallets (UPI, RFID, NFC).
3. Blockchain for Data Security  
Blockchain-based access control logs for tamper-proof vehicle entry records, secure data sharing between authorities and private security systems.
4. AI & Deep Learning Integration  
Implement deep learning models (e.g., CNNs, YOLO, Faster R-CNN) for highly accurate license plate recognition (ANPR) and vehicle identification, AI-based anomaly detection to identify unauthorized

vehicles, fake license plates, or suspicious activity.

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